Role of Expectation on Perceived Time of Visual Events

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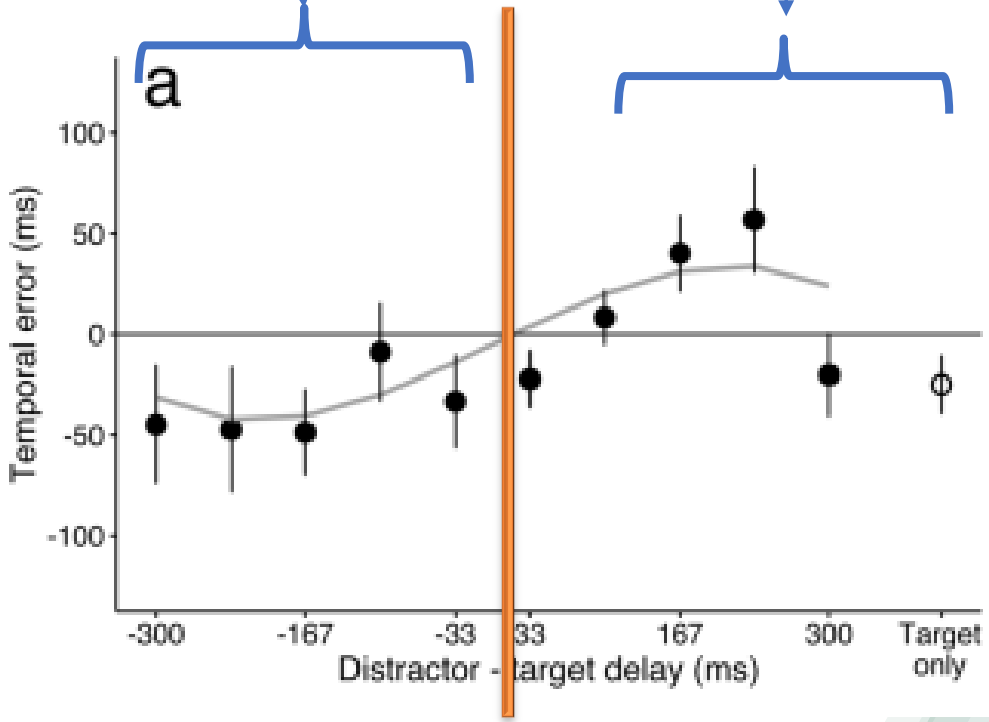
Temporal Context Affects Perceived Time of Visual Events

irrelevant events

Distractor appeared **before** Target

Distractor appeared **after** Target



- **Earlier** distractors caused the **target** to appear **earlier**
- **Later** distractors caused to **later** perception of the **target event**
- This effect was consistent when the target was unknown

Jovanovic & Mamassian (2020)

Research Question

How perceived time affected by distractor events? (**Effect of attention**)

Can **expectation** change perceived moment of event?

What happens when expectation is

Met?

Unmet?

Hypothesis

The perceived time of target event would get **attracted towards** presentation time of "**distractor**"

- Jovanovic & Mamassian (2020)

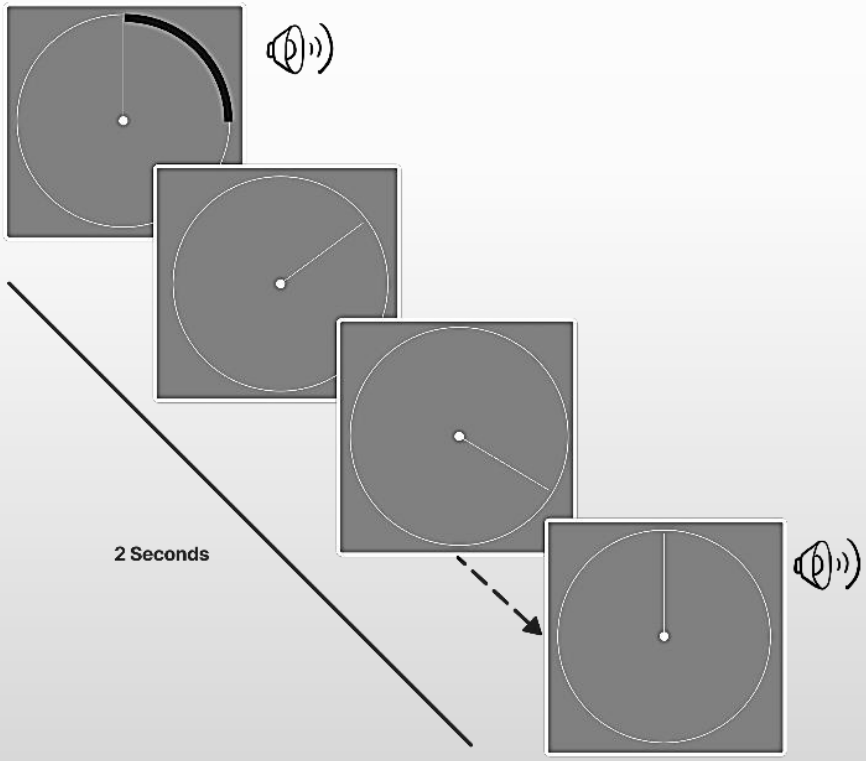
Met expectations will cause in earlier perception of target.

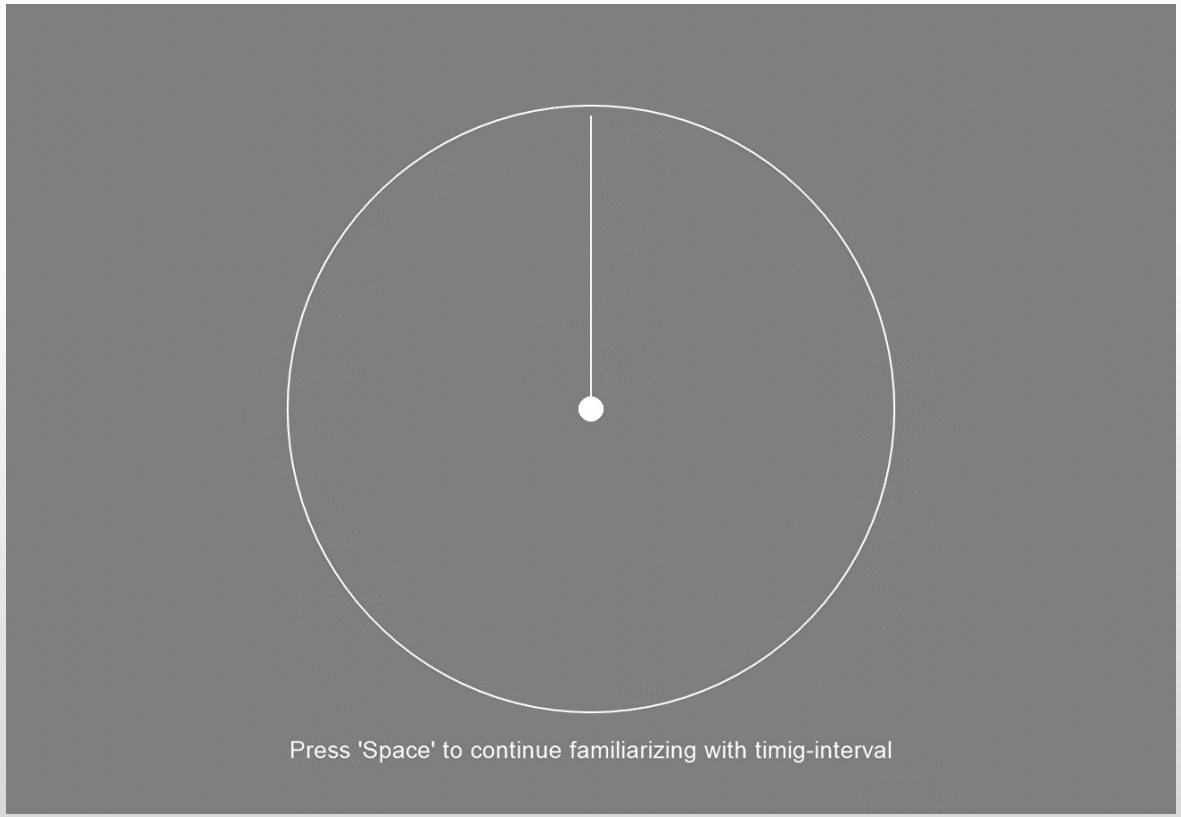
- Kok, Jehee, & De Lange (2012)

Unmet expectations would result in a **later** perceived time of event.

- Urgen & Boyaci (2019)

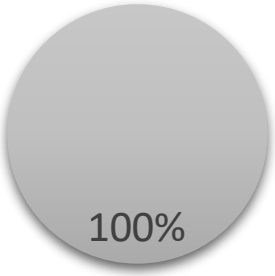
Familiarization





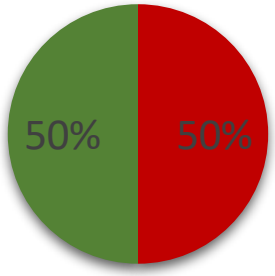
Procedure

Baseline



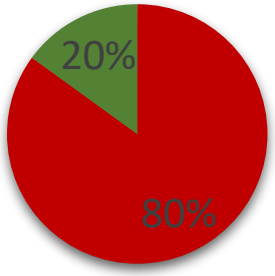
Target

No Expectation

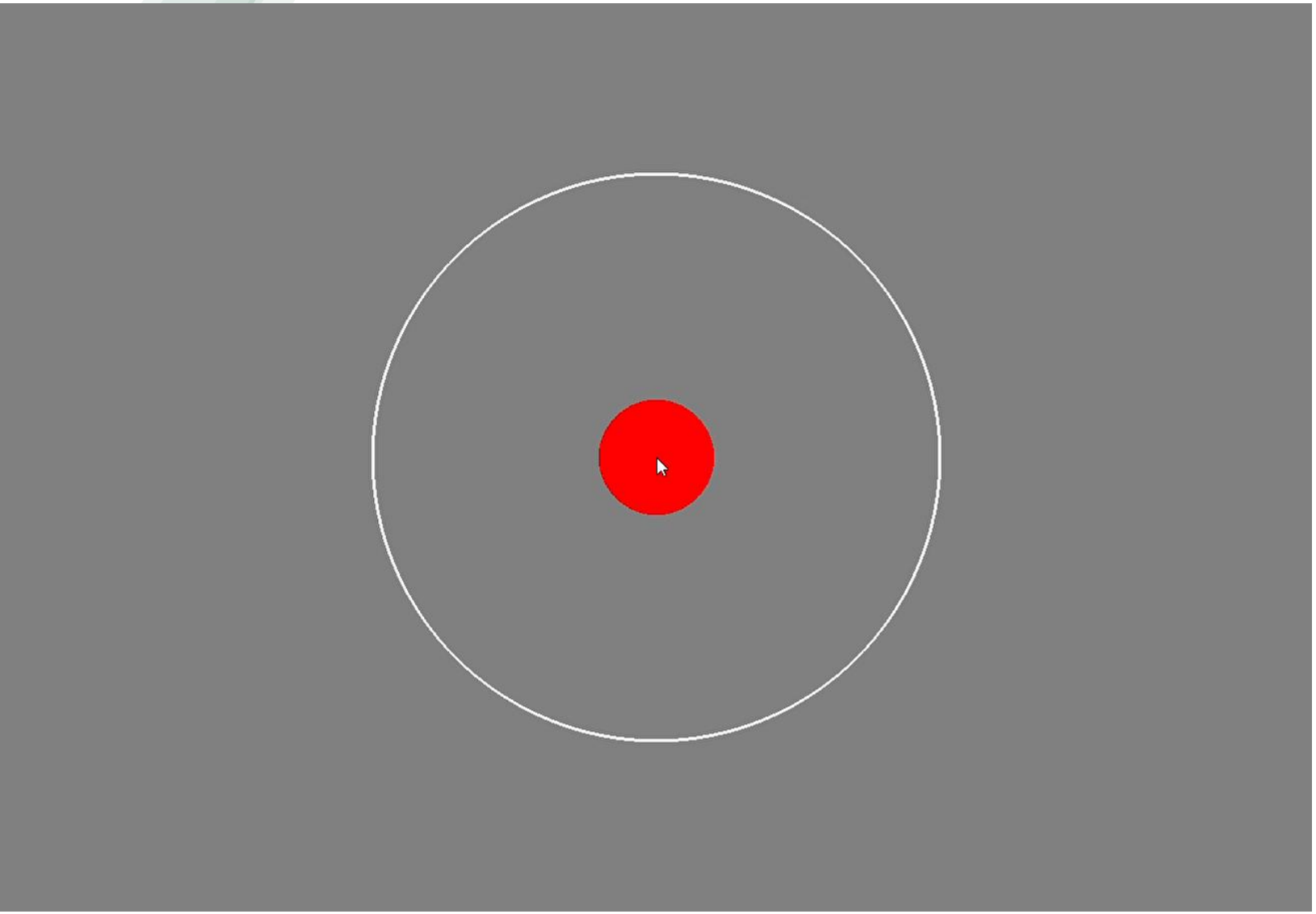


Target Distractor

Expectation



Target Distractor



Baseline

1. No distractor
2. Started and Finished with BEEP!

Results on baseline

1. Baseline condition: Single stimuli condition

$$Perceived\ Time = \alpha * \sin(\pi\, t_t) + \beta \cdot x$$

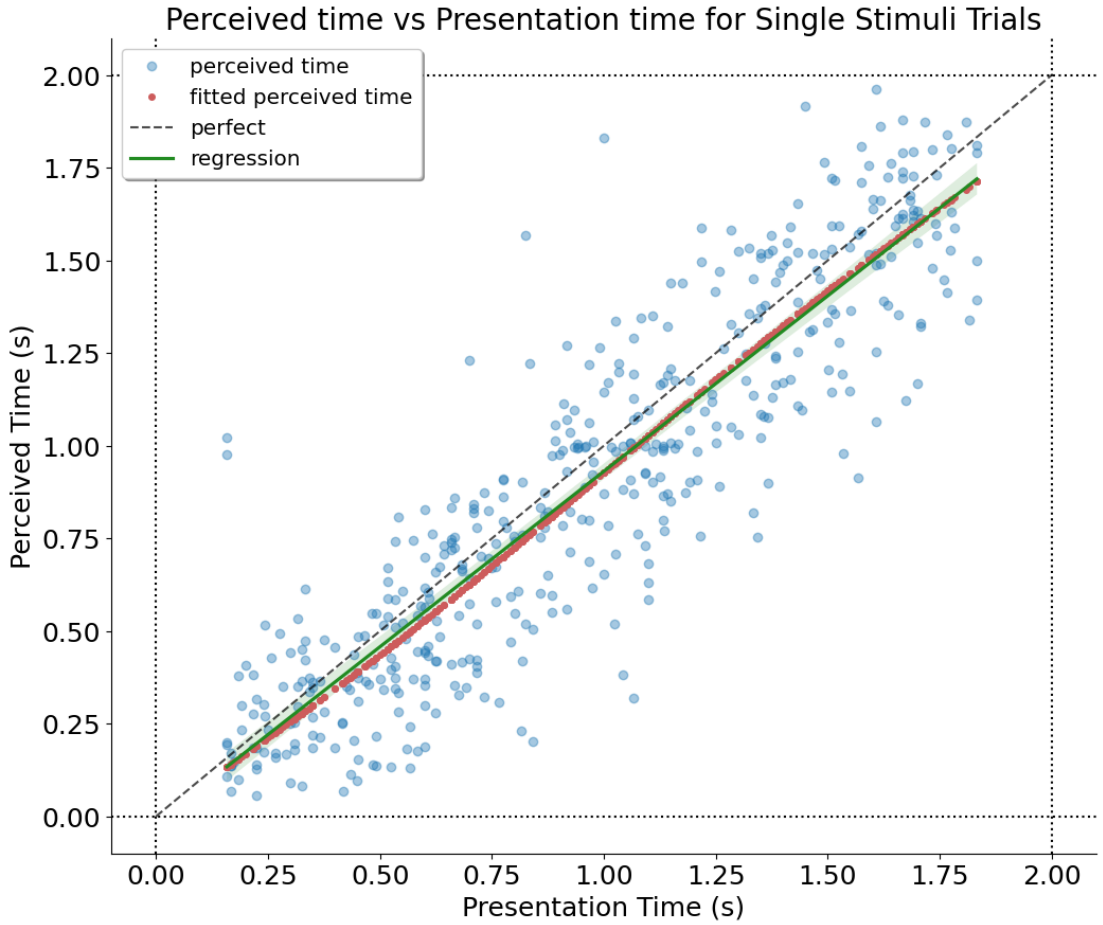


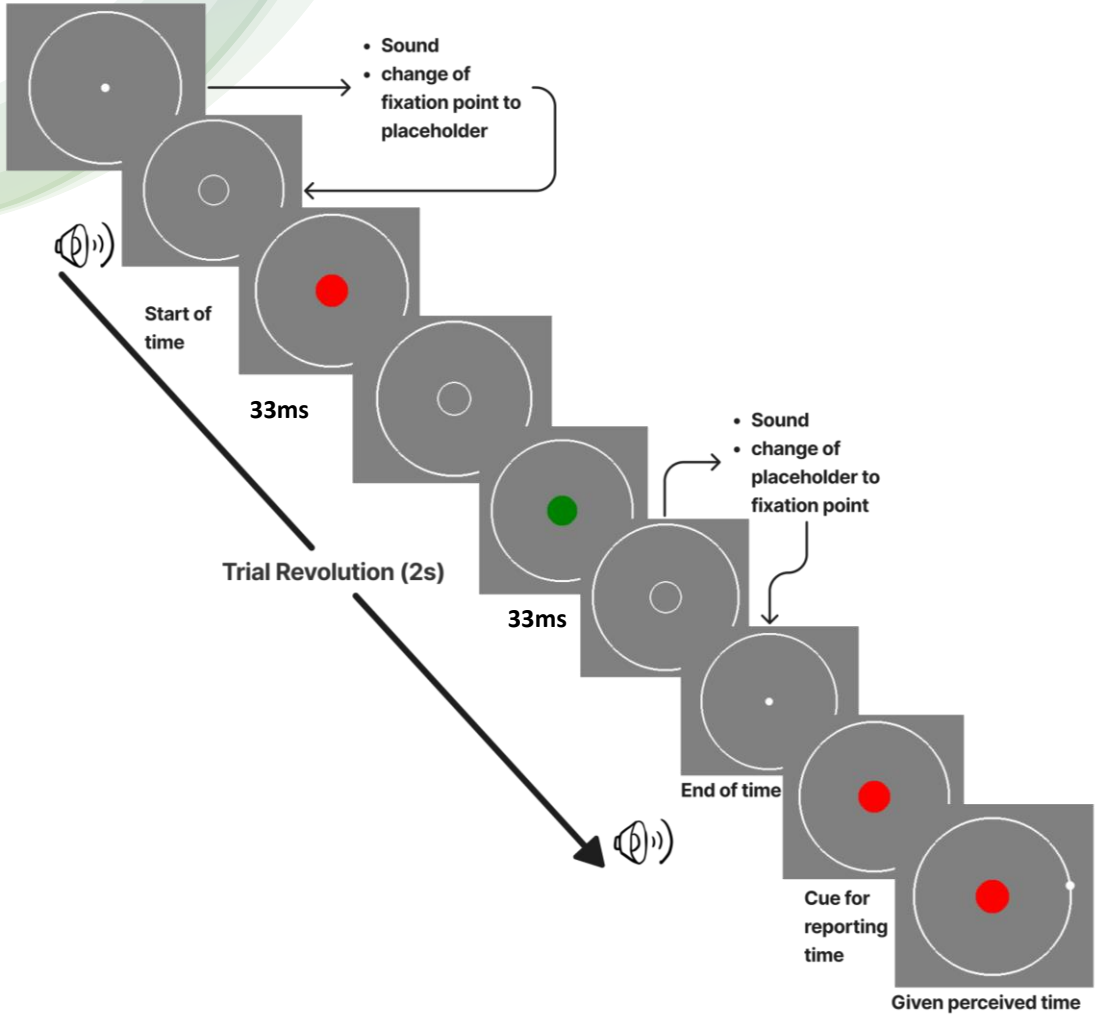
Figure 4: Perceived time vs Presentation time for Single Stimuli Trials.

Blue dots represent the perceived time

Red dots represent the fitted perceived time.

Green line represents the regression line.

Test Trials



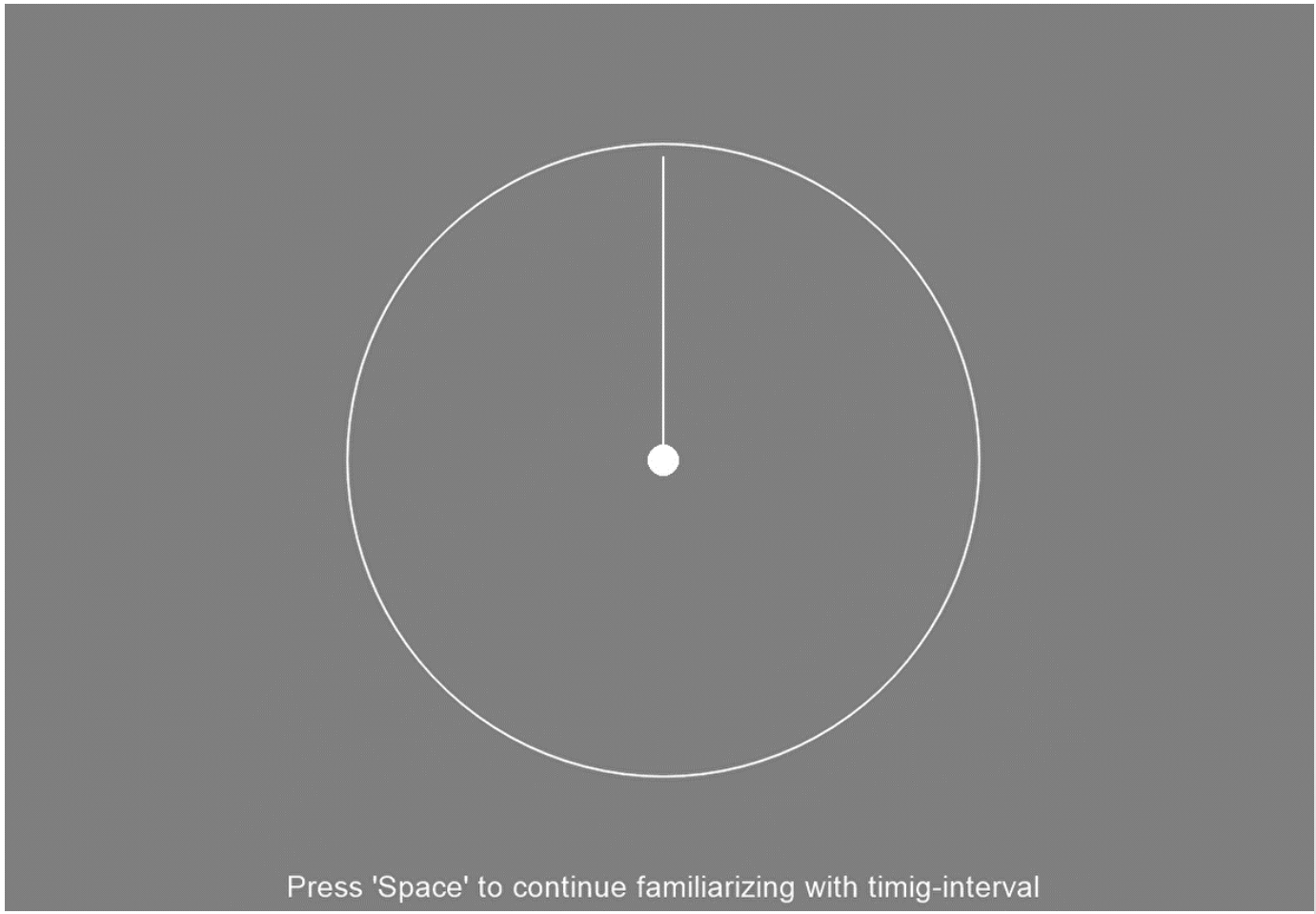
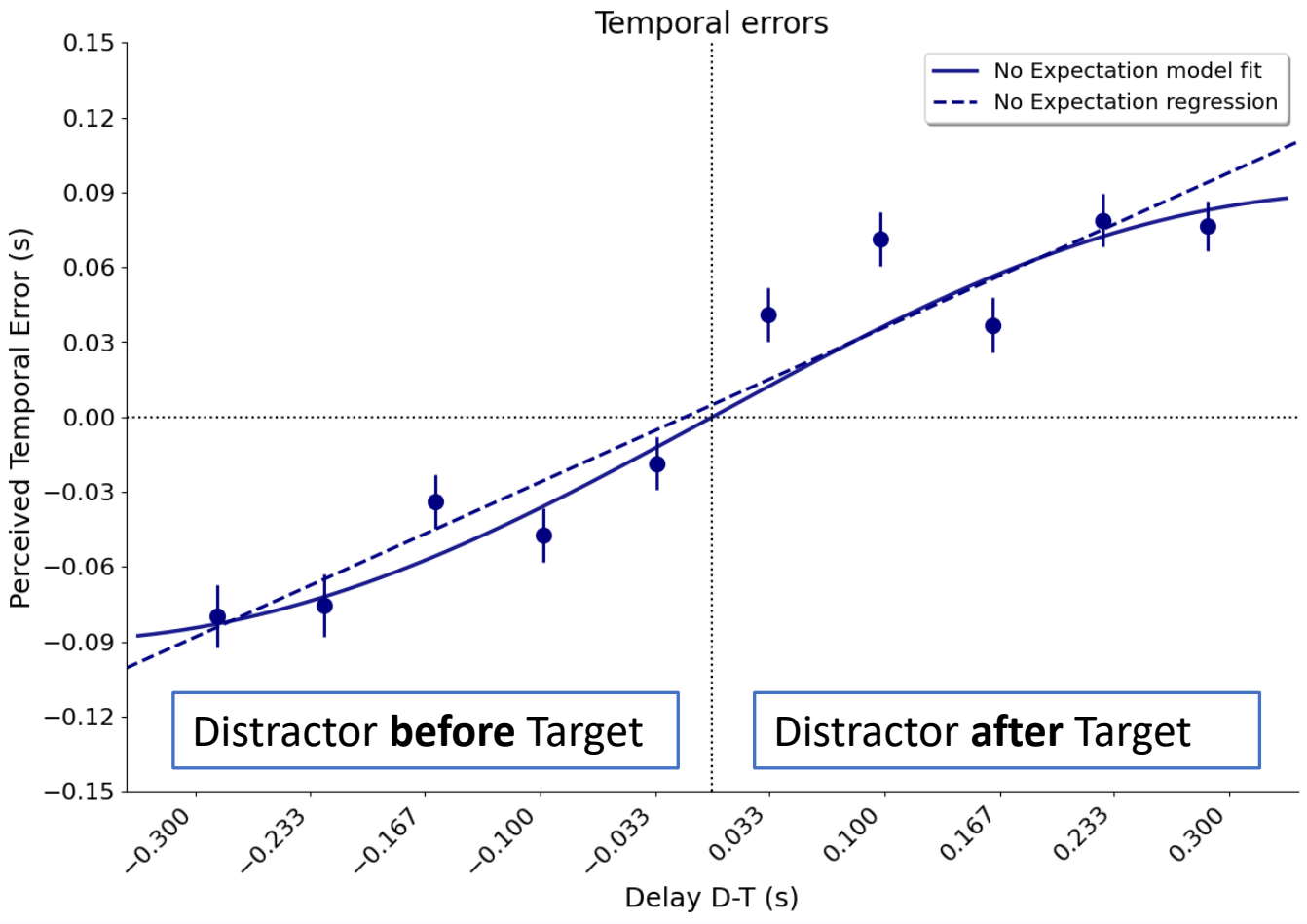


Figure : Demonstration of an experimental trial timeline

2. No-Expectation Condition: Effect of time between stimuli

$$Temporal\ Error = f(x) = t_t' - t_t = \alpha \cdot \frac{(t_d - t_t)}{\sigma} \cdot \exp\left[-\frac{(t_d - t_t)^2}{\sigma^2}\right]$$



- Significant main effect of Delay

  β = 0.310, SE = 0.0151, t = 20.5, **p < .0001**

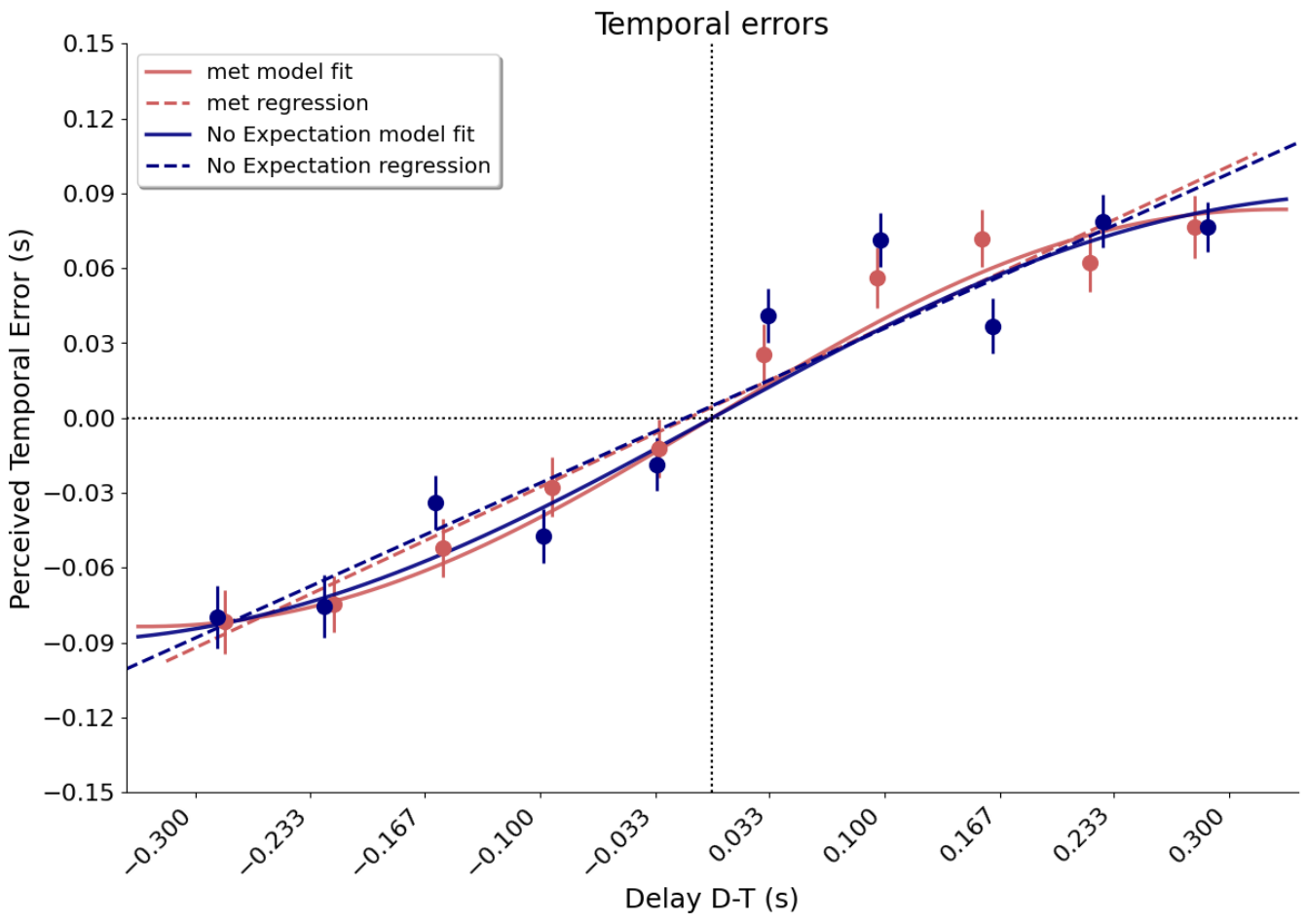
- Main effect of derivative of gaussian function of Delay (Distribution of Delay)

  β = 0.196, SE = 0.094, t = 2.08, **p = 0.0371**

HA_1: The perceived time of target event would get **attracted towards** presentation time of "**distractor**"

3. When there is an expectation towards a certain color

$$Temporal\ Error = f(x) = t_t' - t_t = \alpha \cdot \frac{(t_d - t_t)}{\sigma} \cdot \exp\left[-\frac{(t_d - t_t)^2}{\sigma^2}\right]$$

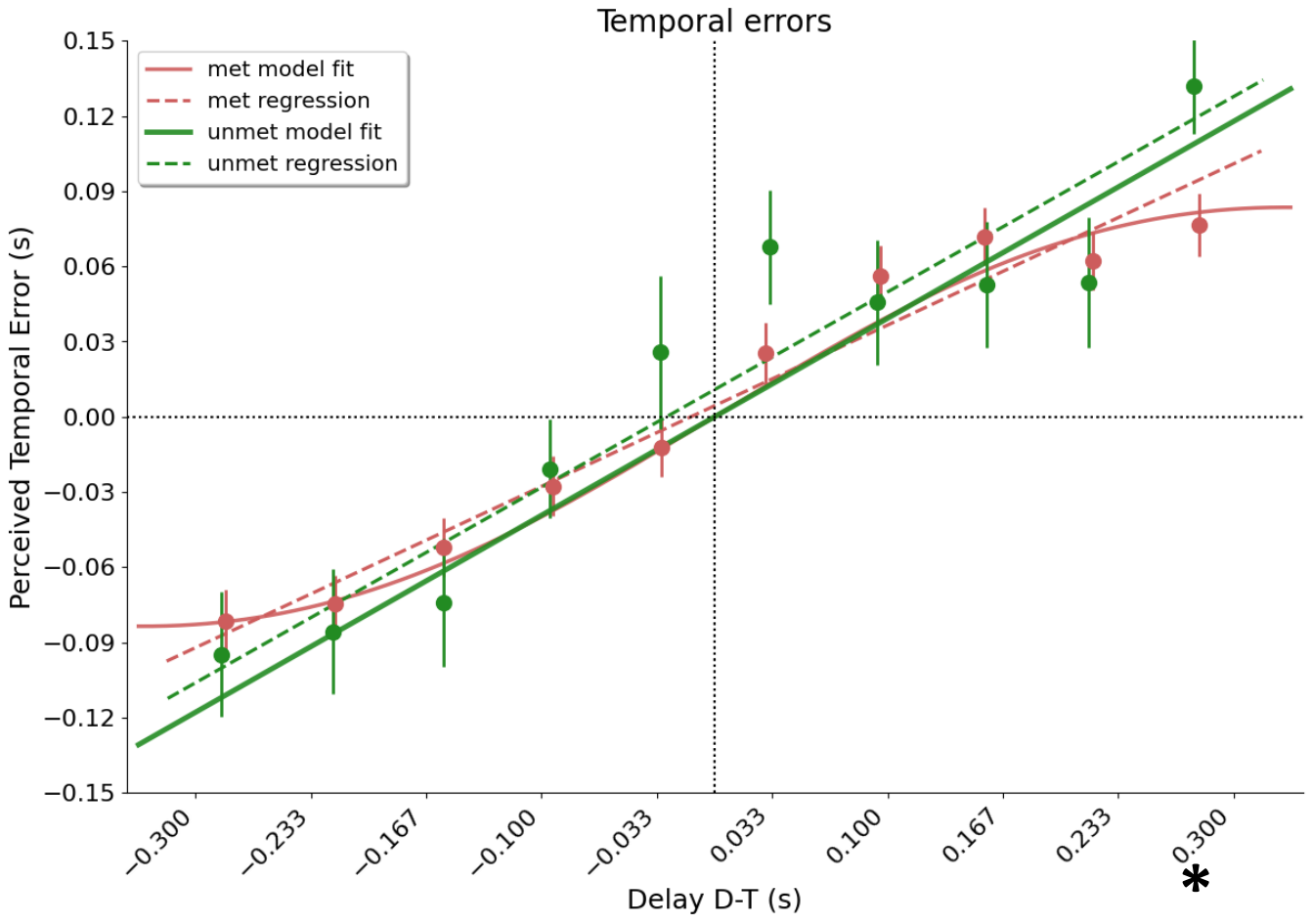


- No main effect of expectation

HA_2: When there is an **expectation** about the upcoming stimulus, this will result in an **earlier perception** of the onset of the target stimulus in comparison to the actual presentation time.

4. Met expectations vs Unmet Expectations

$$Temporal\ Error = f(x) = t_t' - t_t = \alpha \cdot \frac{(t_d - t_t)}{\sigma} \cdot \exp\left[-\frac{(t_d - t_t)^2}{\sigma^2}\right]$$



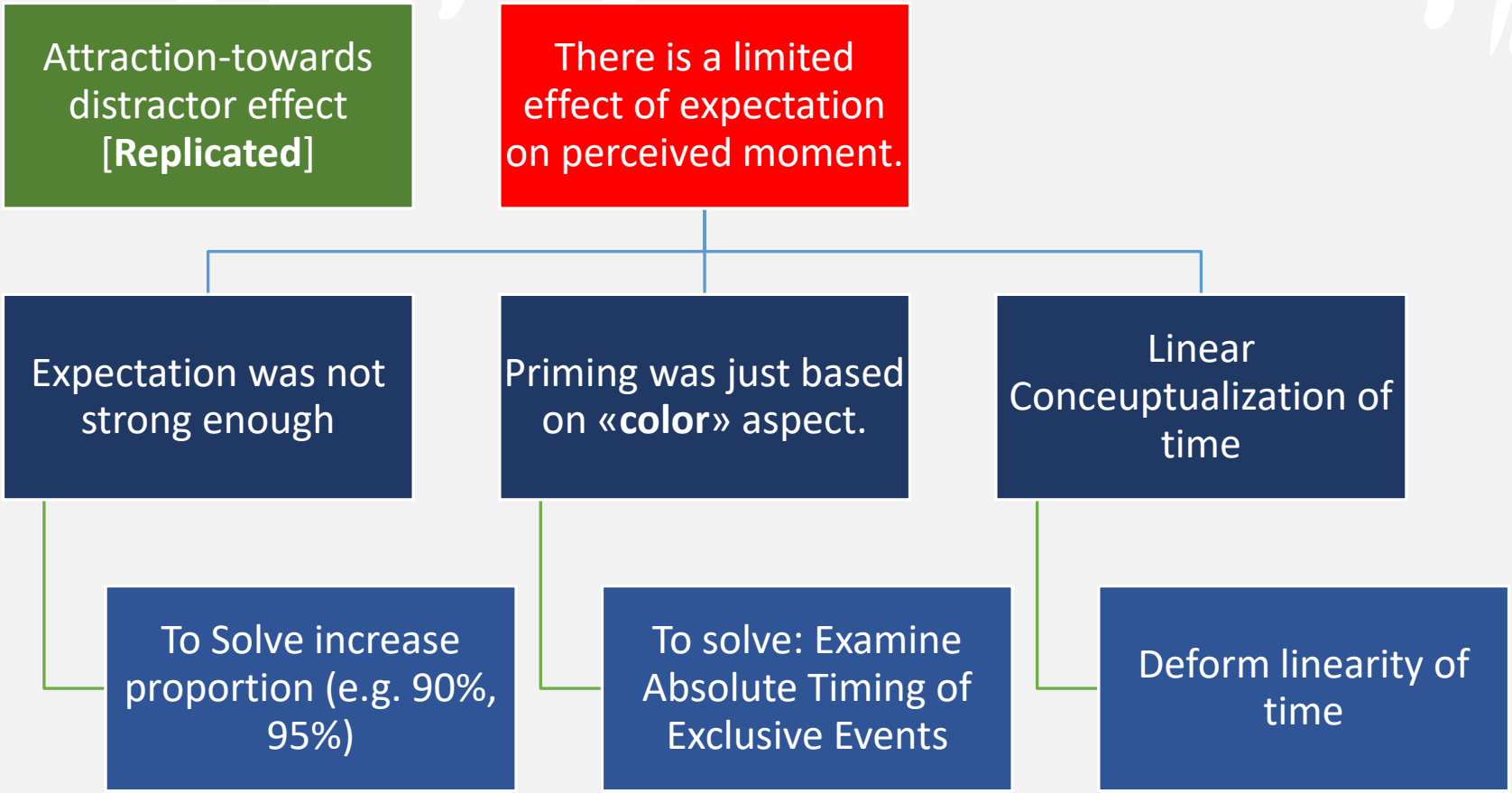
- No main effect of expectation
- **Interaction** Effect

β = 0.07857, SE = 0.03877, t = 2.027, **p = 0.0427**

HA_3: When the expectation is **unmet** we expect participants to perceive the target stimulus later than the actual presentation time, and the error variability to be greater.

*:t=-2.53,p=0.016

Discussion


Attraction-towards distractor effect [**Replicated**]
There is a limited effect of expectation on perceived moment.
Expectation was not strong enough
Priming was just based on «**color**» aspect.
Linear Conceuptualization of time
To Solve increase proportion (e.g. 90%, 95%)
To solve: Examine Absolute Timing of Exclusive Events
Deform linearity of time

Discussion

Attraction-towards distractor effect [**Replicated**]

There is a limited effect of expectation on perceived moment.

To speculate these results:

- Expectation was not srong enough
  - To Solve increase proportion (e.g. 90%, 95%)
- Priming was just based on «**color**» aspect.
  - To solve: Examine Absolute Timing of Exclusive Events
- Linear Conceuptualization of time
  - Deform linearity of time

4. Detailed investigation on each of the delay bins

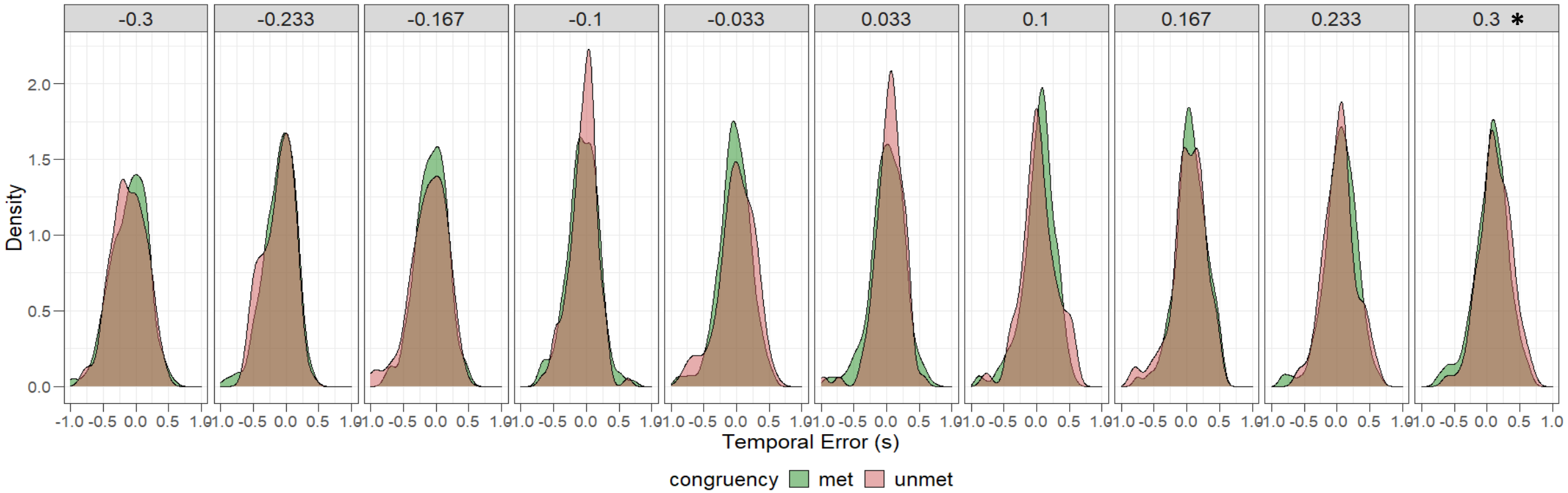


Figure 7: Density distributions of perceived time for each delay bin in the met and unmet conditions

*:t=-2.53,p=0.016

5. Response Time and Reaction Times

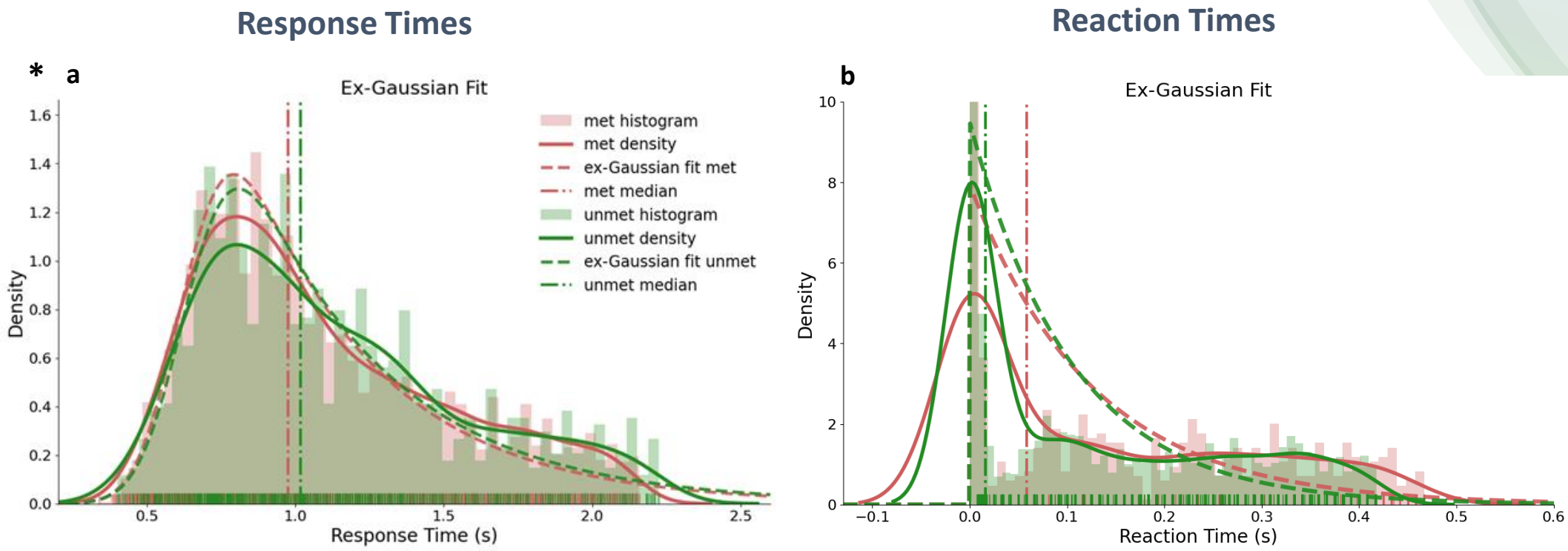


Figure 8: Ex-Gaussian fit for response times **(a)** and for reaction times **(b).** Both of these graphs includes probability density fits (-), histograms, median lines (-.), and ex-gaussian fit(-.) lines for met(green) and unmet (red) trials.

*:t=-2.54, p=0.011

Time

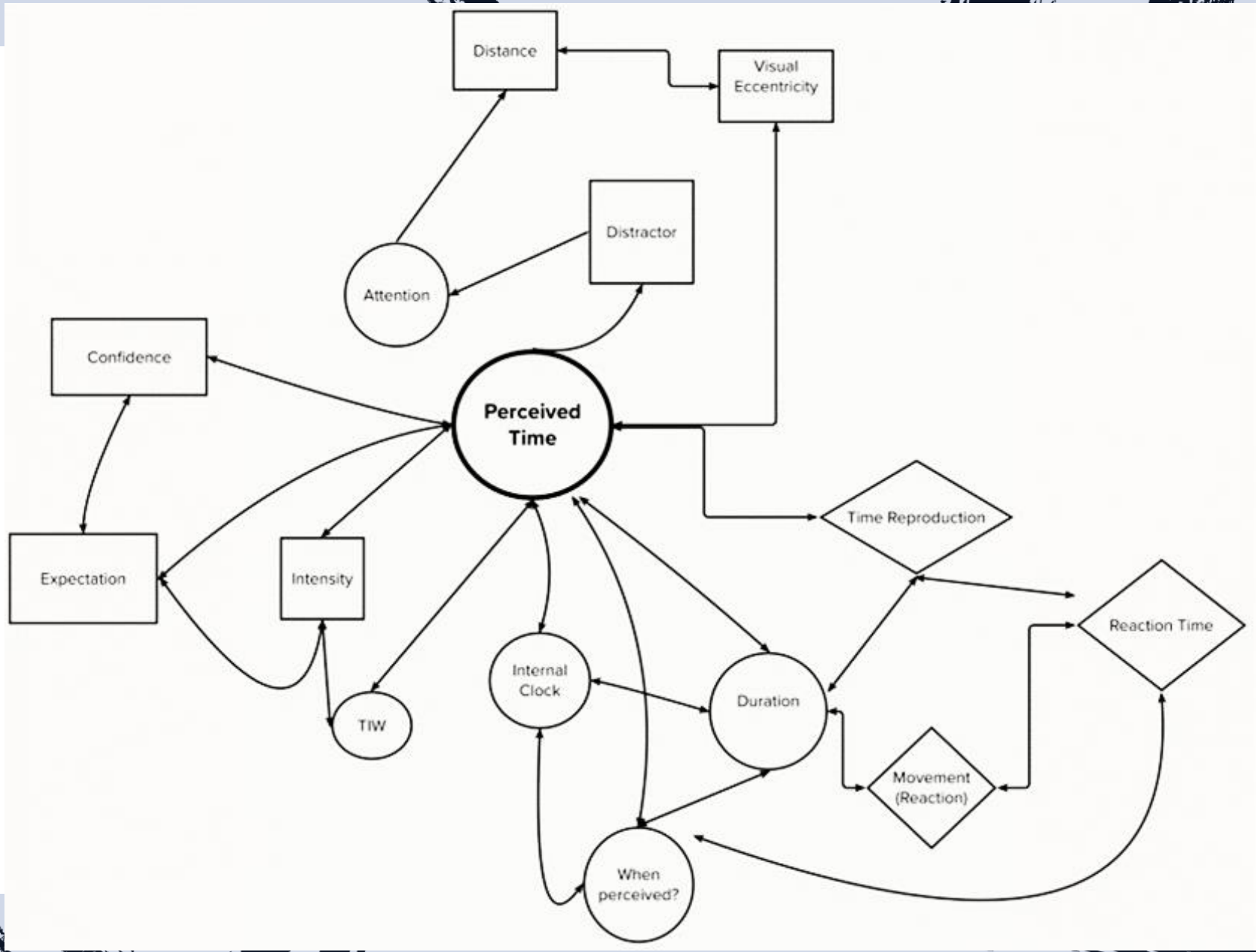
- Fundemental cognitive process
- Active
- Non-isolated

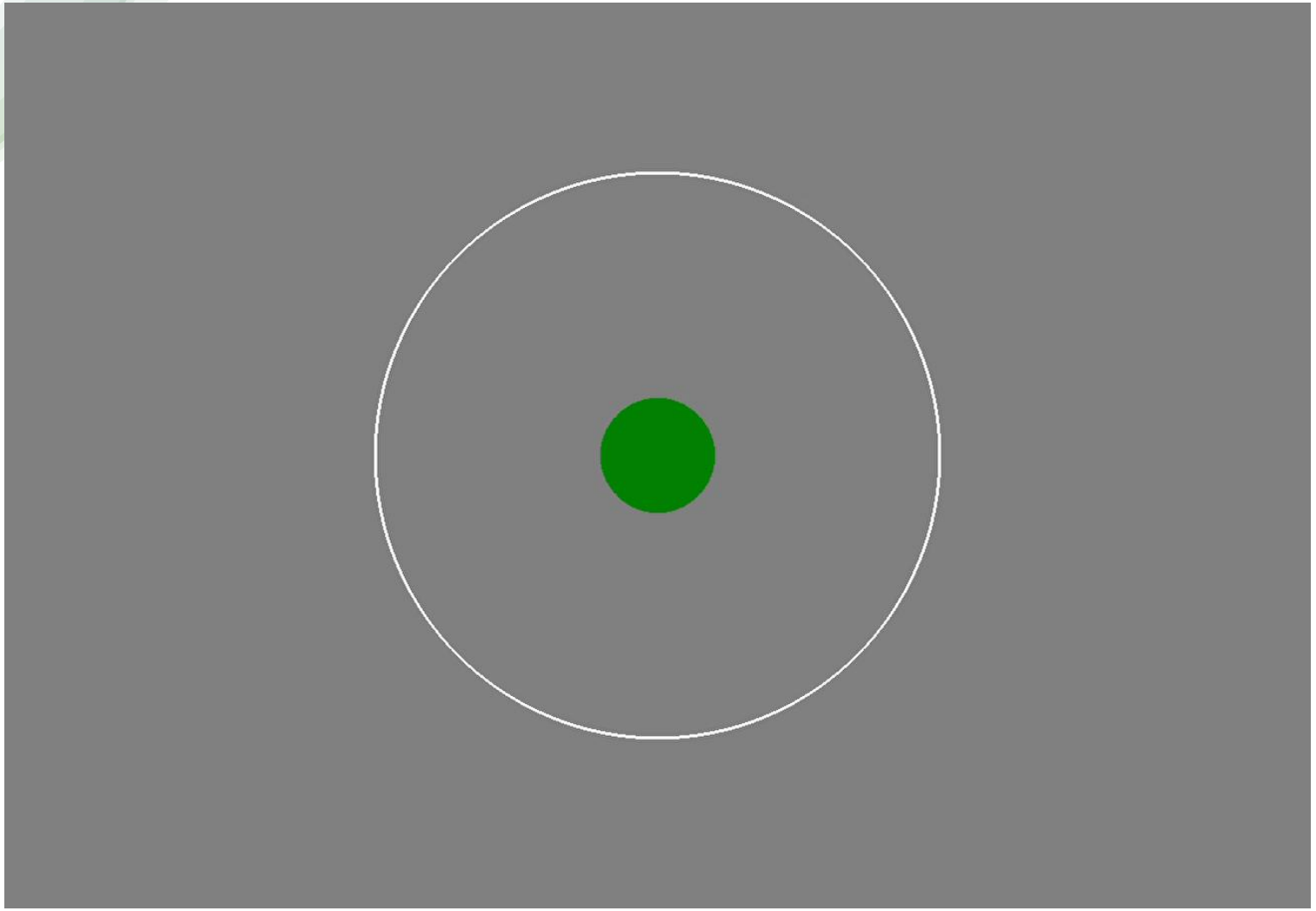
"Time **flies** when you are having fun ."

"Time is perceived as shorter when humans are **engaged** in some absorbing activity, because the important factor is that **attention is distracted** from time."

Ref: Tse, Intriligator, Rivest & Cavanagh, 2004; Seifried & Ulrich, 2011; Birngruber, & Ulrich, 2014

Contextual Effects

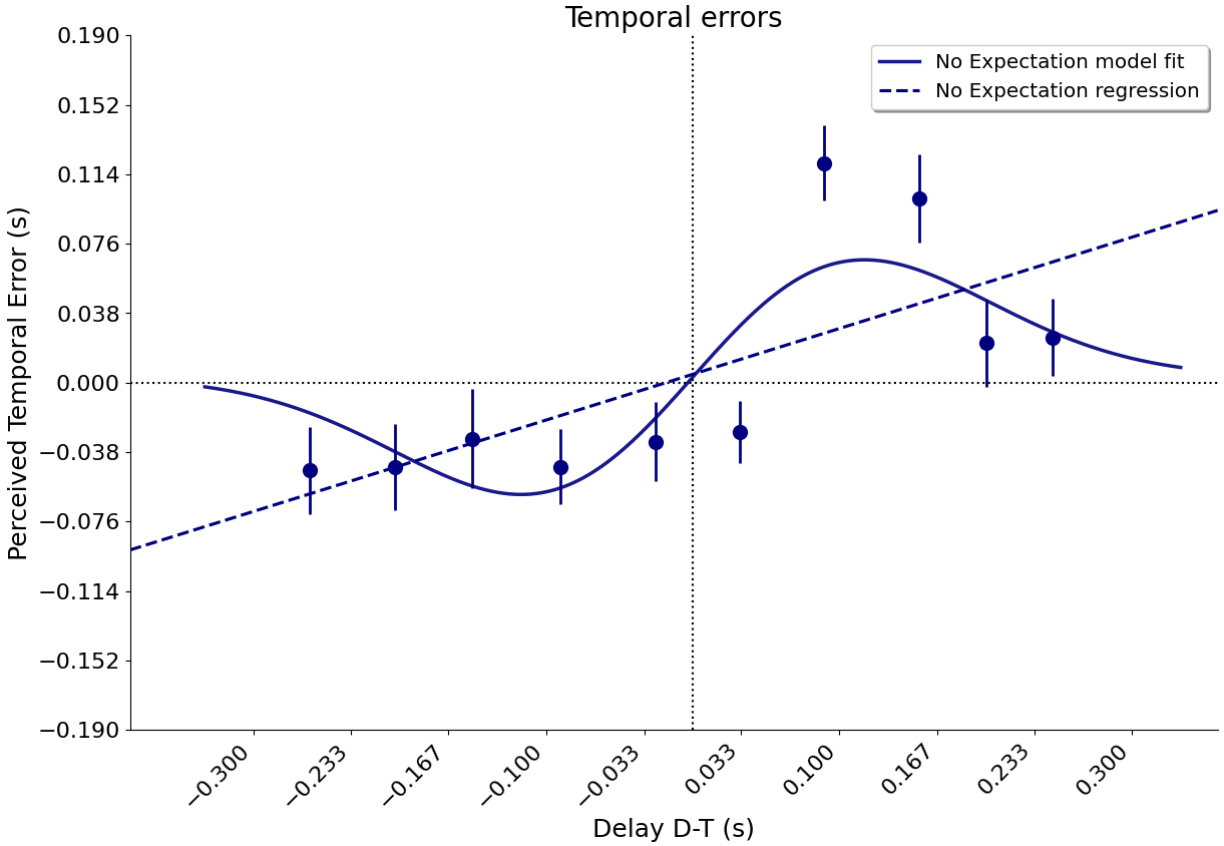


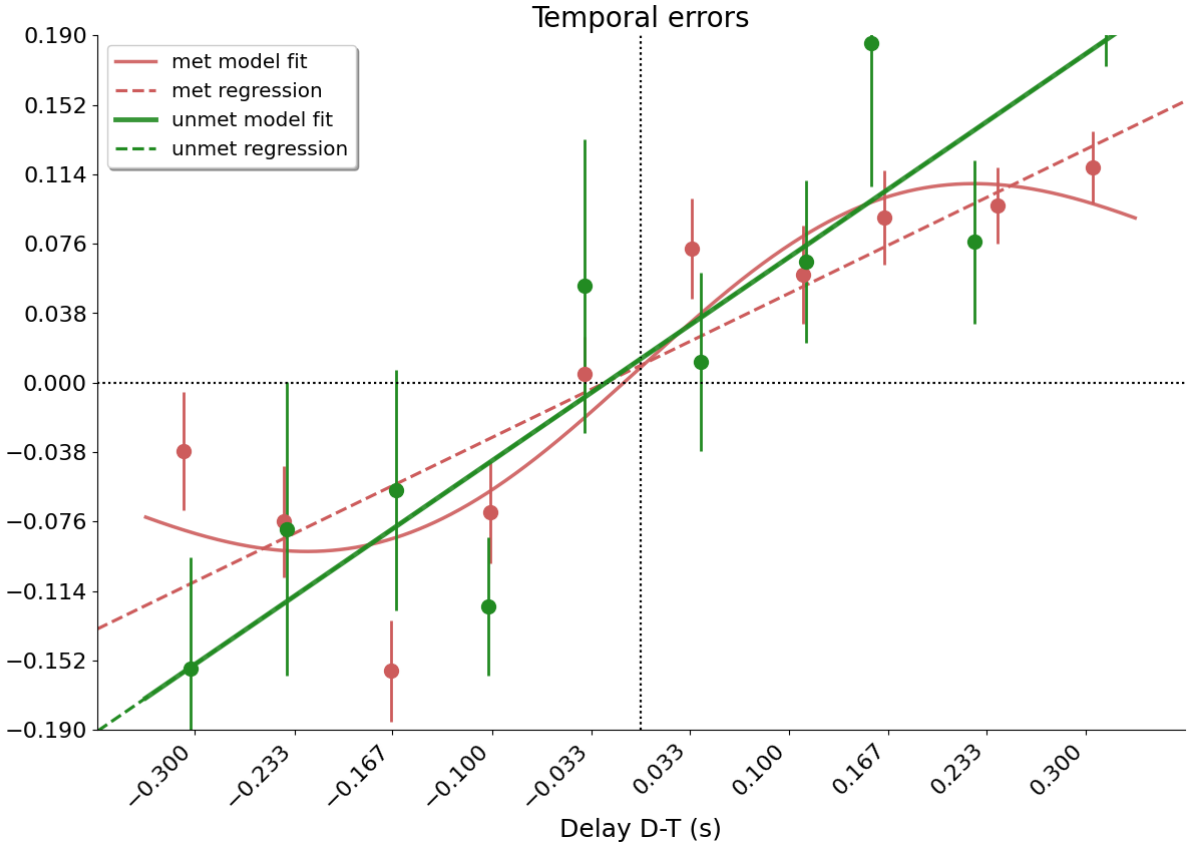


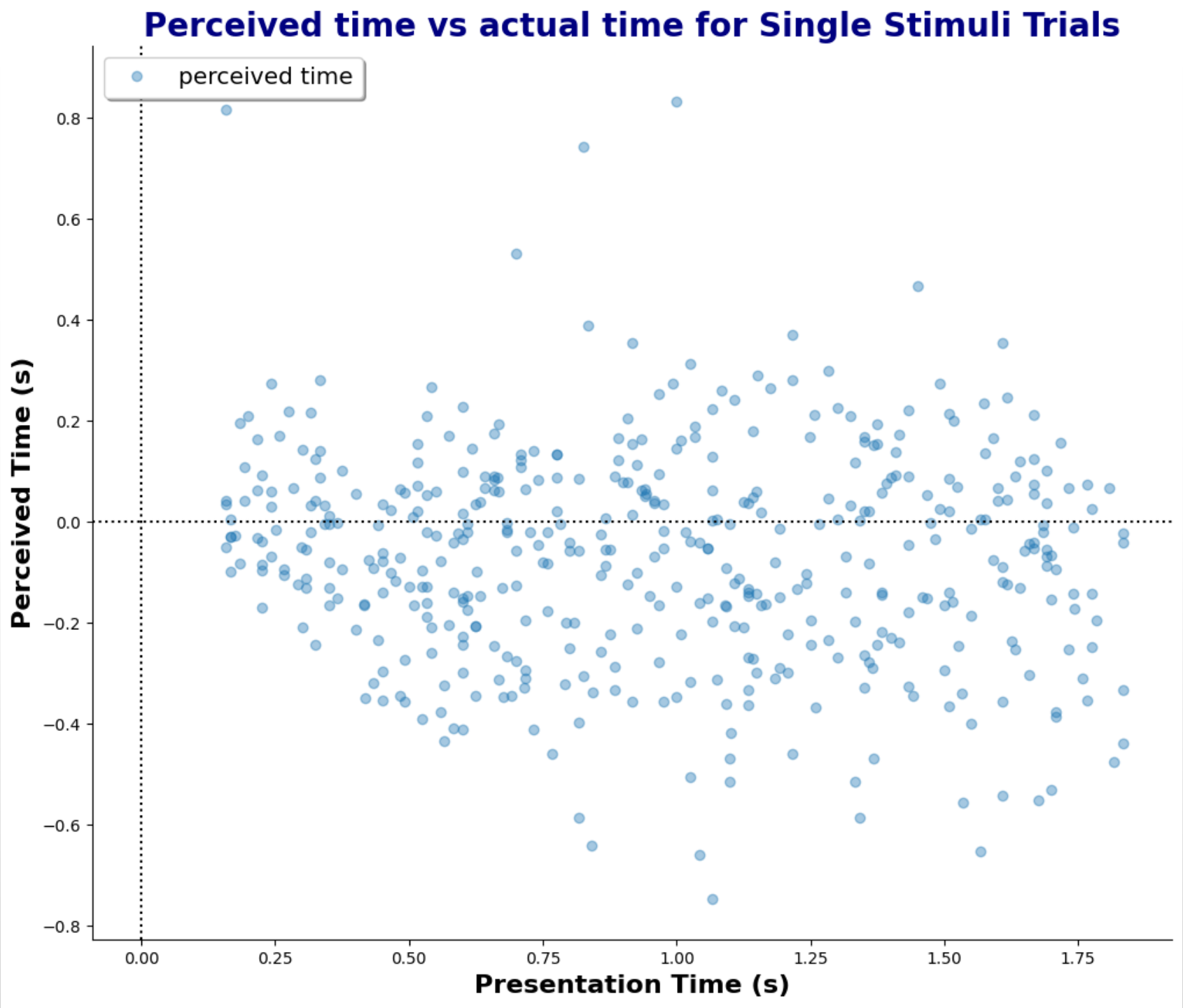
Training

1. Single Stimulus Training with Feedback
2. Single Stimulus Training without Feedback (Baseline)
3. Double Stimulus Training without Feedback

Subject-wise Analysis






Perceived time vs actual time for Single Stimuli Trials
perceived time
Perceived Time (s)
Presentation Time (s)
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75